Final Project Report

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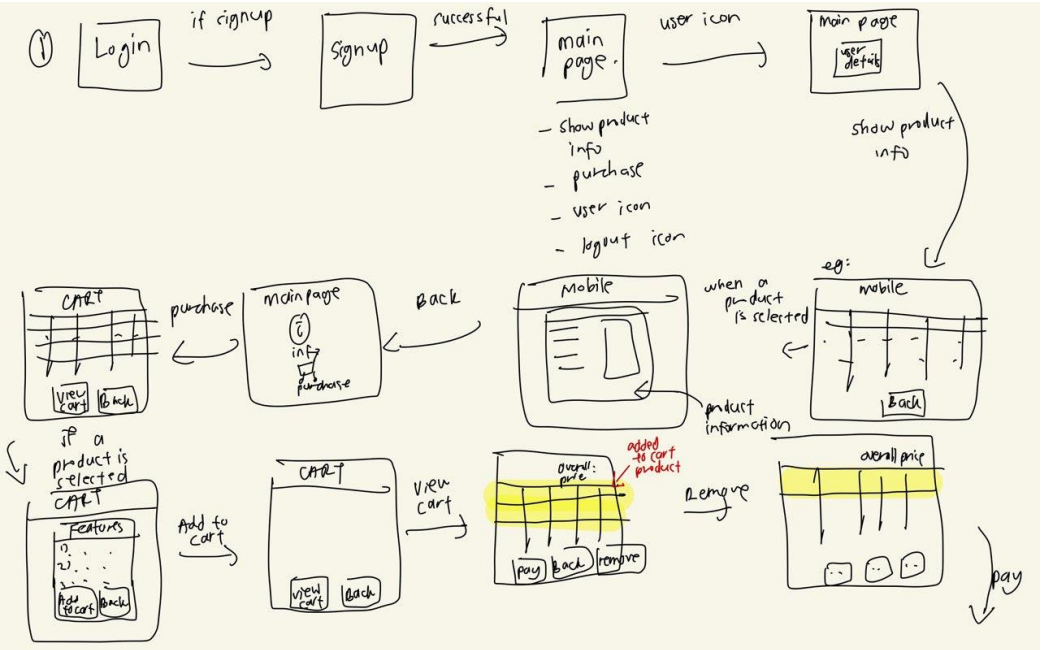
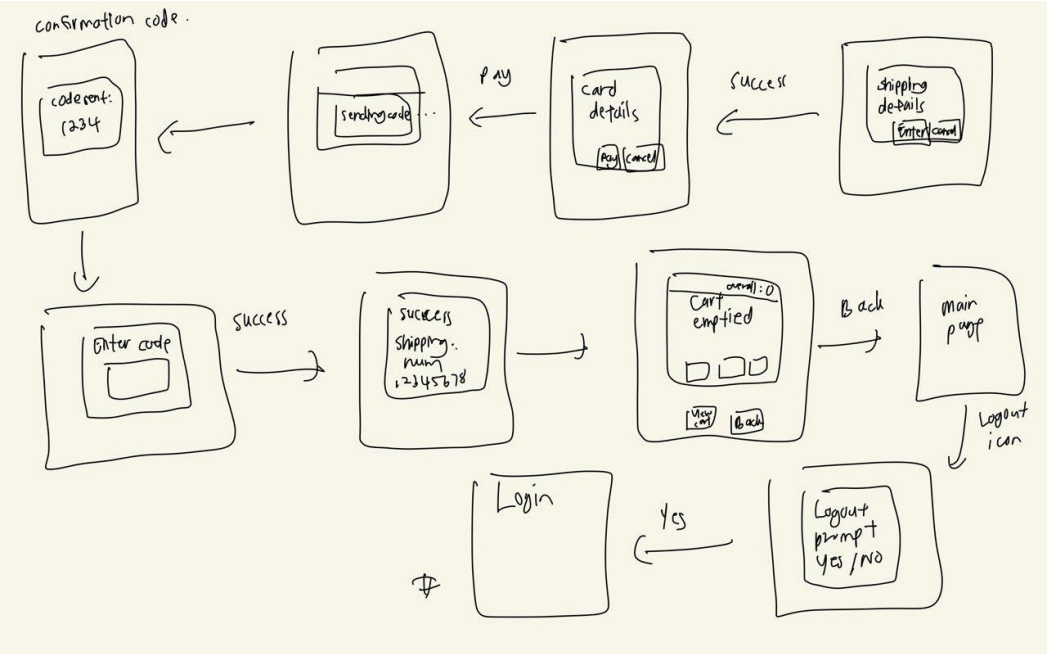
1. Briefly describe the project purpose:
The purpose of this project is to create an application where user can view the product informations and checkout the products that they are interested in. My project is an electronic-based shop.

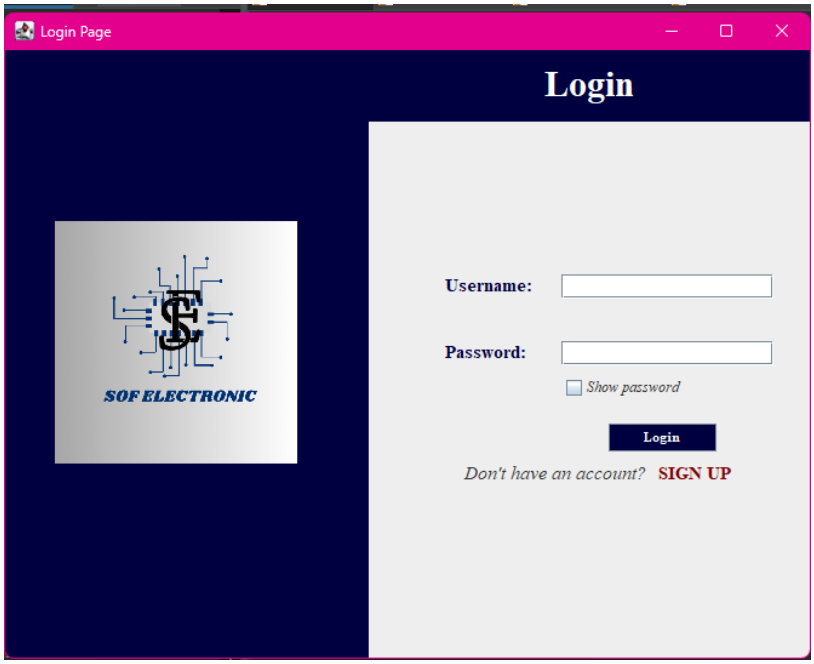
2. Draw the logic flow of the program (with flowchart):



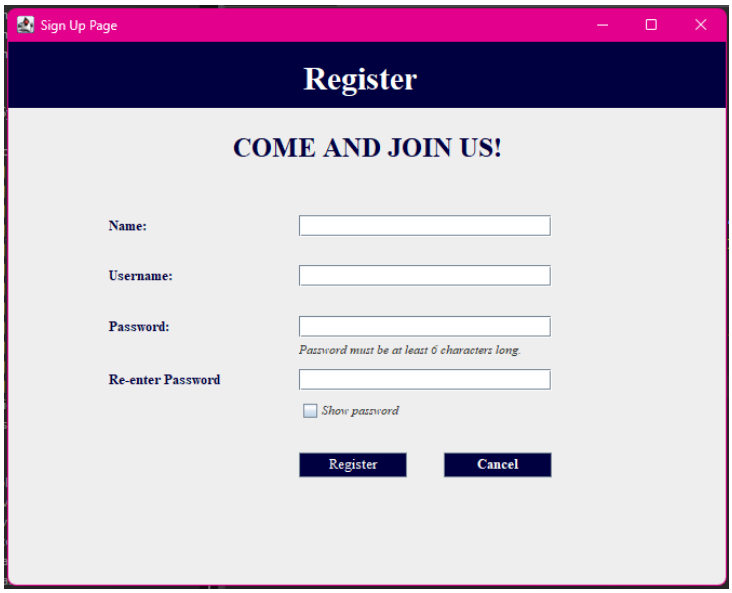


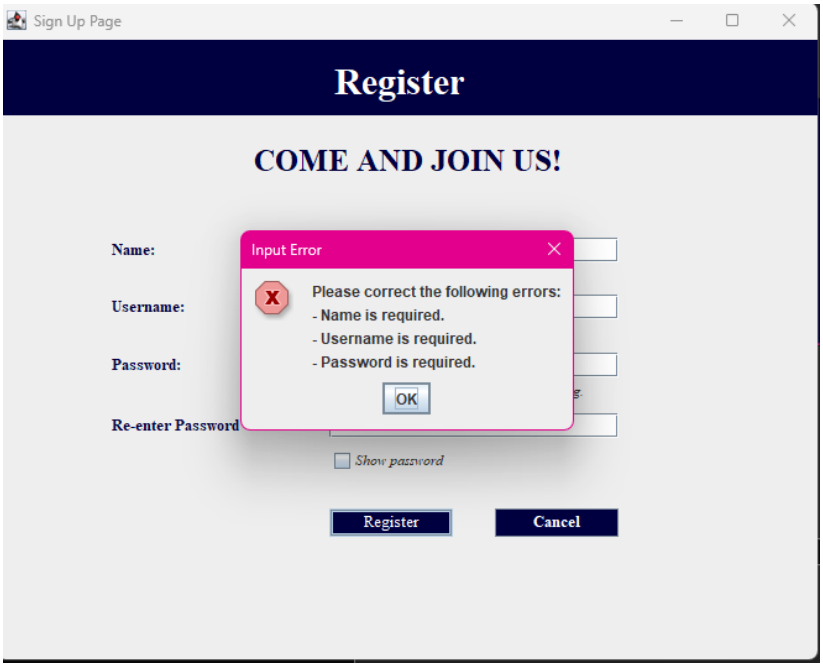
3. Provide screenshots for each screen with brief description:

1)The first page is Login page where it prompts for user's username and password.



2) When you click sign up, it redirected to this register page. Non-existing users can register using unique usernames. If username existed, it will show warning message. If the fields are incorrectly filled or be left empty when register button is clicked, it will show strings of error messages. Successfuly registered users are saved into users text file.






```
Full Name:Rujhan
Username:rujhan03
Password:rujhan123

Full Name:Ameera
Username:myaisk
Password:ameera123

Full Name:Penny Lee
Username:penny123
Password:penelope

Full Name:Sofea Jane
Username:pengkyu
Password:sofea123
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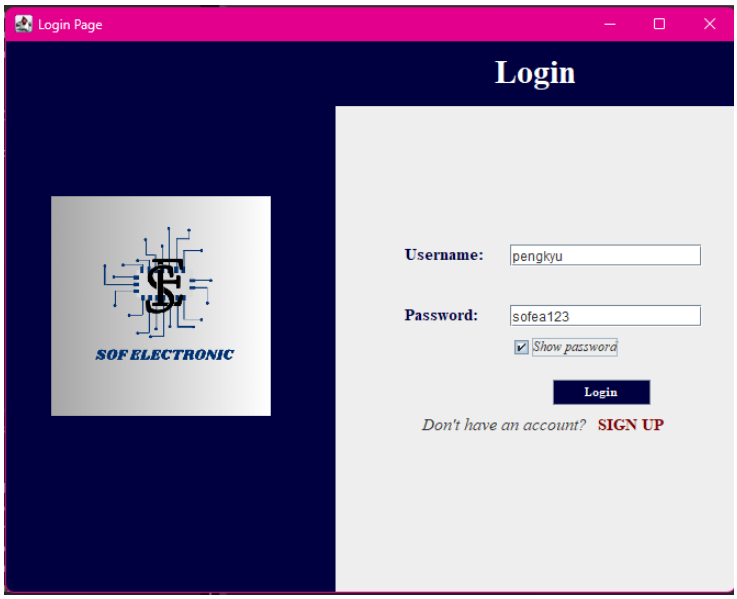
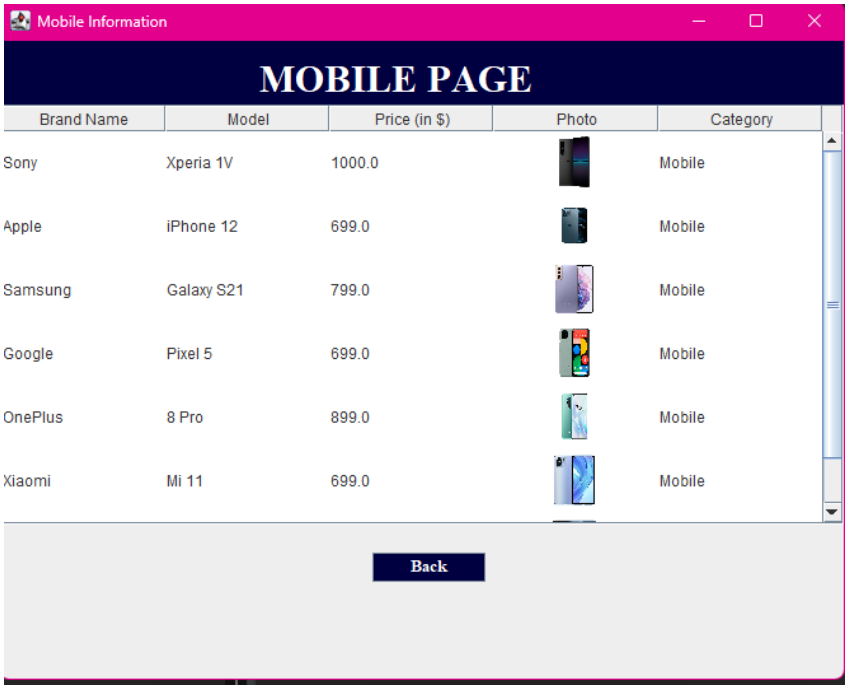
3) Back to the login page, this is the example of corect user's input. Password is showed when user clicks on 'Show password' checkbox.

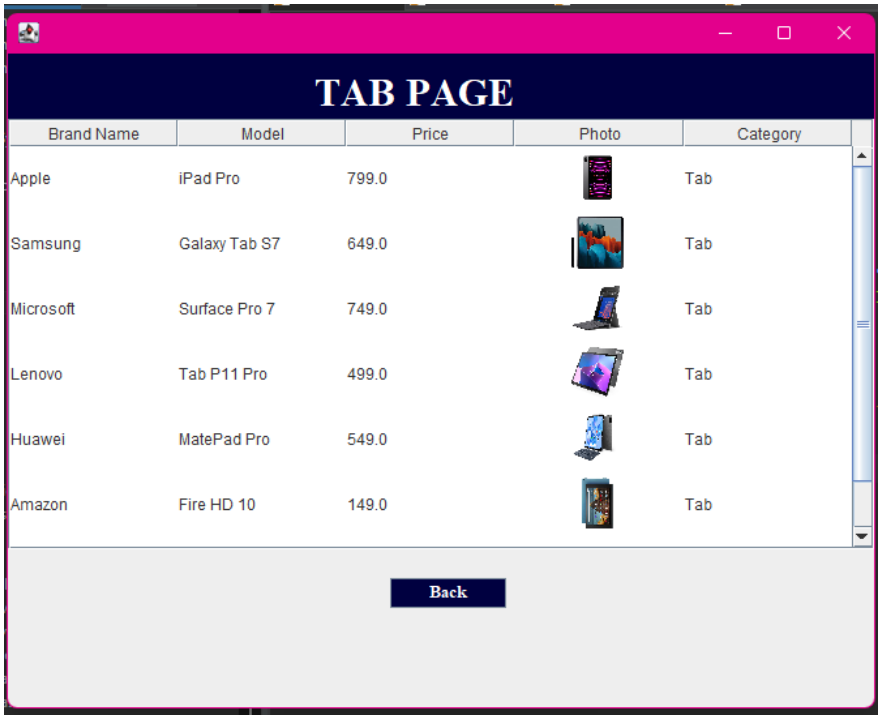


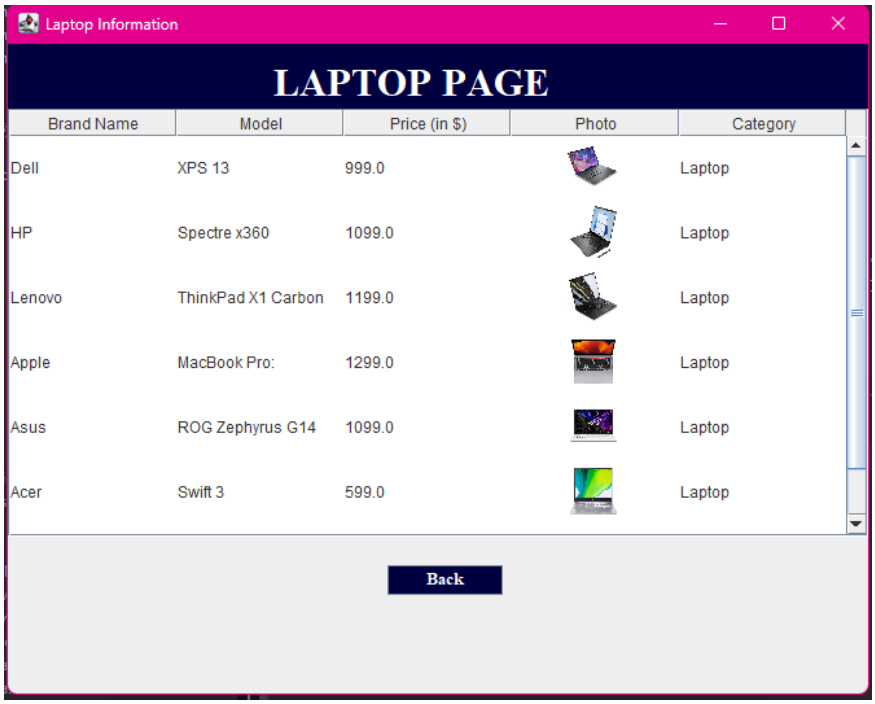
4) After a successful login, you will be directed to main page where there are user's icon, logout icon, show product informations and make purchases

5) When user's icon is clicked, it shows the user's name and username

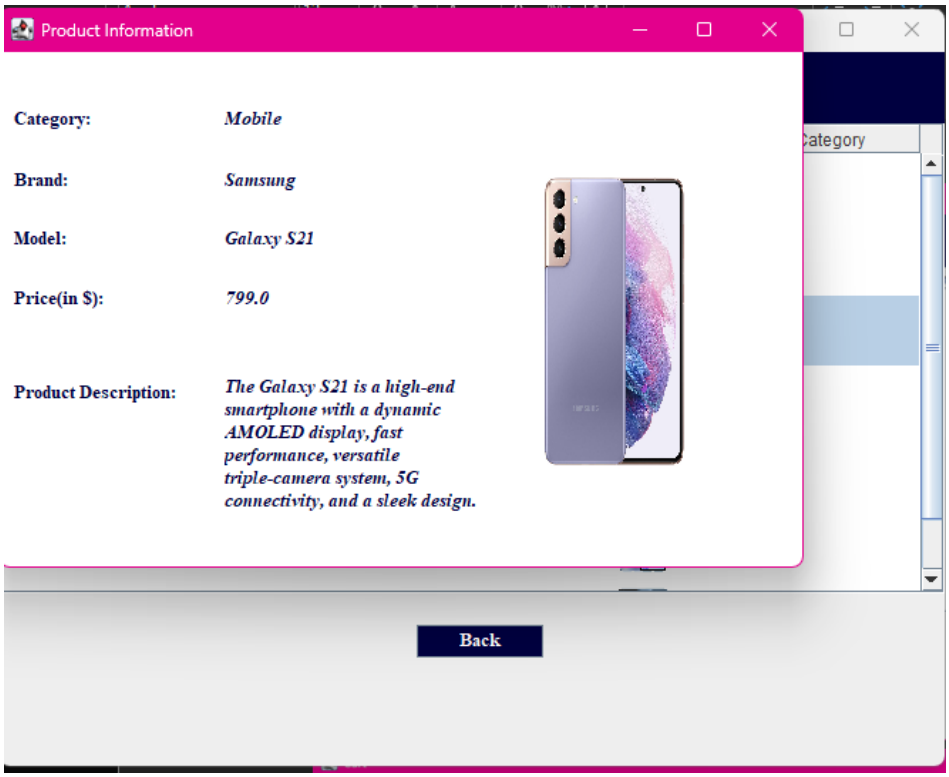
6) When you click on show products, it will show this page containing product informations, so these are three frames of three products




TAB PAGE
Brand Name
Model
Price
Photo
Category
Apple
iPad Pro
799.0
Tab
Samsung
Galaxy Tab S7
649.0
Tab
Microsoft
Surface Pro 7
749.0
Tab
Lenovo
Tab P11 Pro
499.0
Tab
Huawei
MatePad Pro
549.0
Tab
Amazon
Fire HD 10
149.0
Tab
Back

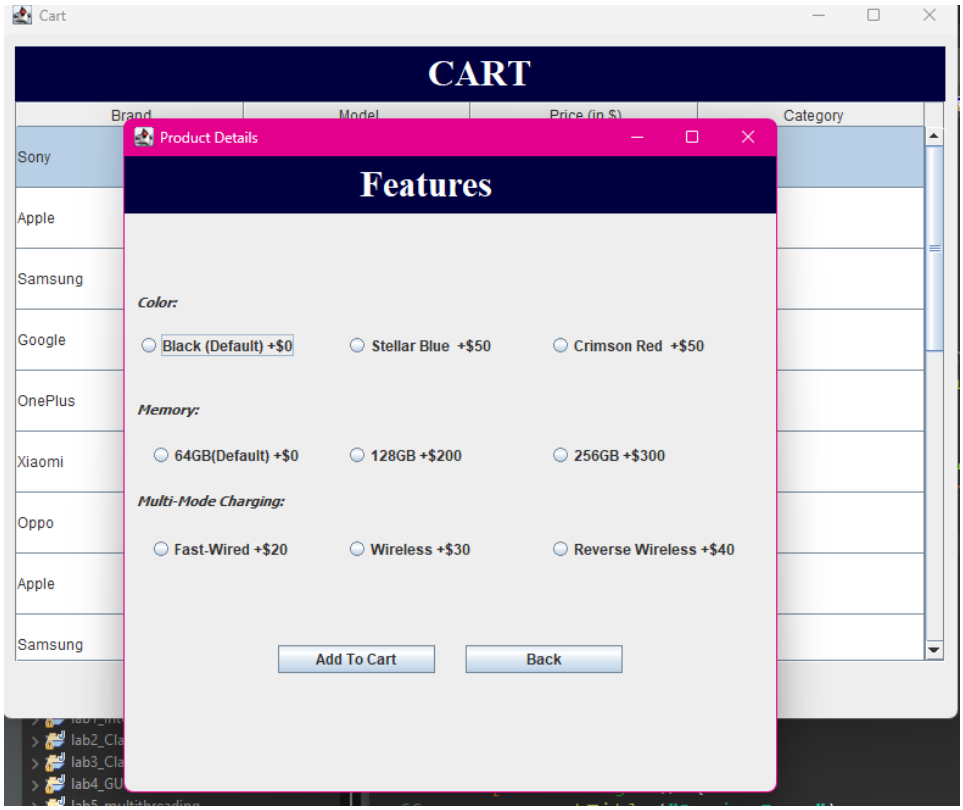

Laptop Information
LAPTOP PAGE
Brand Name
Model
Price (in $)
Photo
Category
Dell
XPS 13
999.0
Laptop
HP
Spectre x360
1099.0
Laptop
Lenovo
ThinkPad X1 Carbon
1199.0
Laptop
Apple
MacBook Pro:
1299.0
Laptop
Asus
ROG Zephyrus G14
1099.0
Laptop
Acer
Swift 3
599.0
Laptop
Back

7) When you clicked on one of the product, it will show you the product details such as below

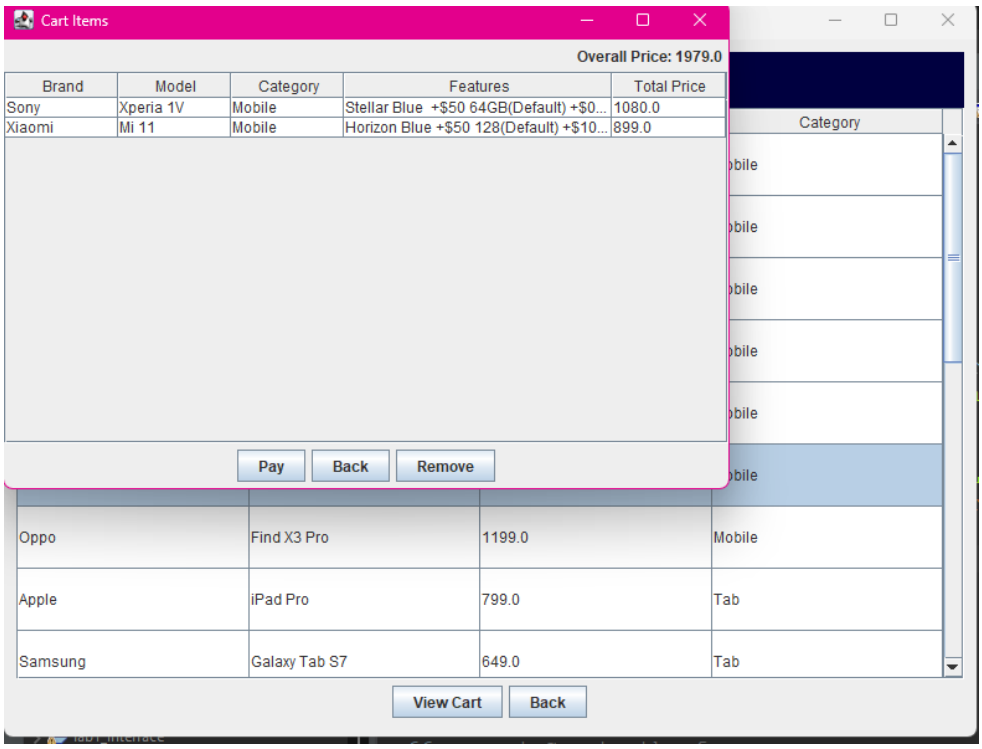


8) Next, when you click purchase, you will be directed to the cart frame where it has list of all the products in store

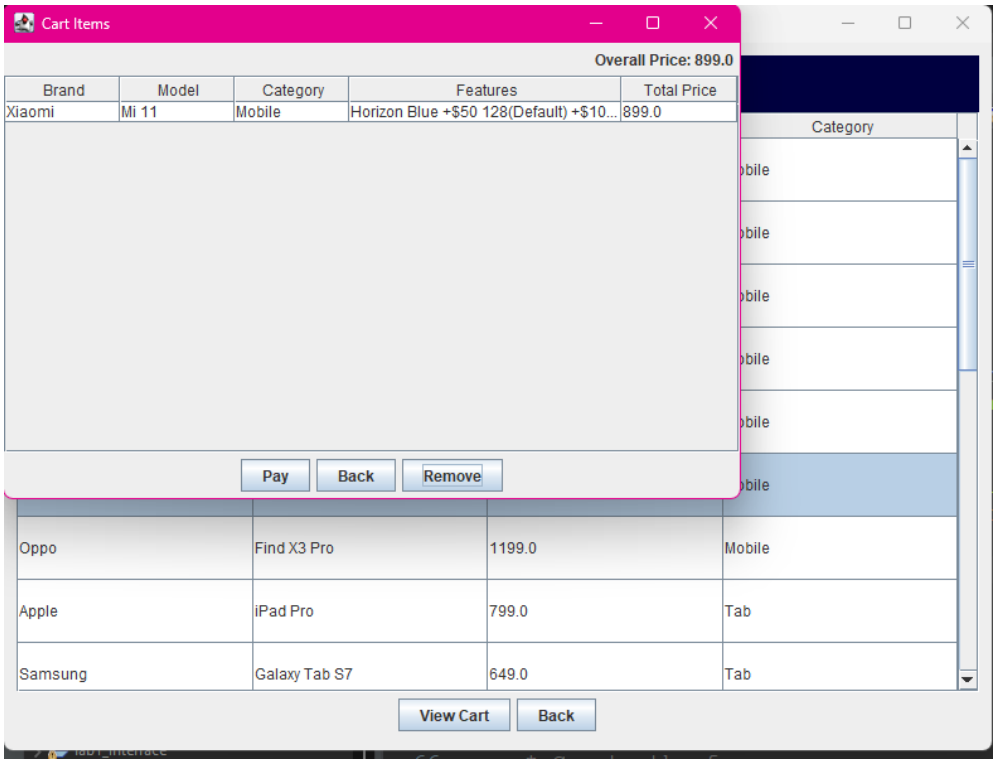
9) When you click on a product, it will show you the features and additional fees for each of the product



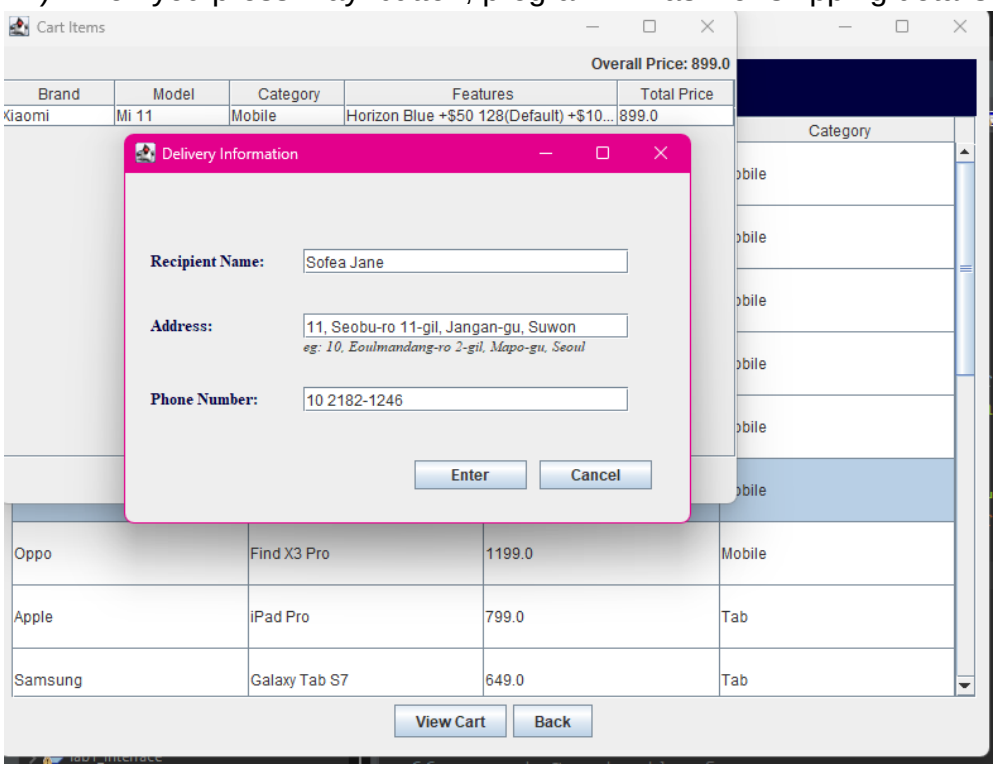
10) When you click on 'View Cart' button, it will show the products you had added to cart



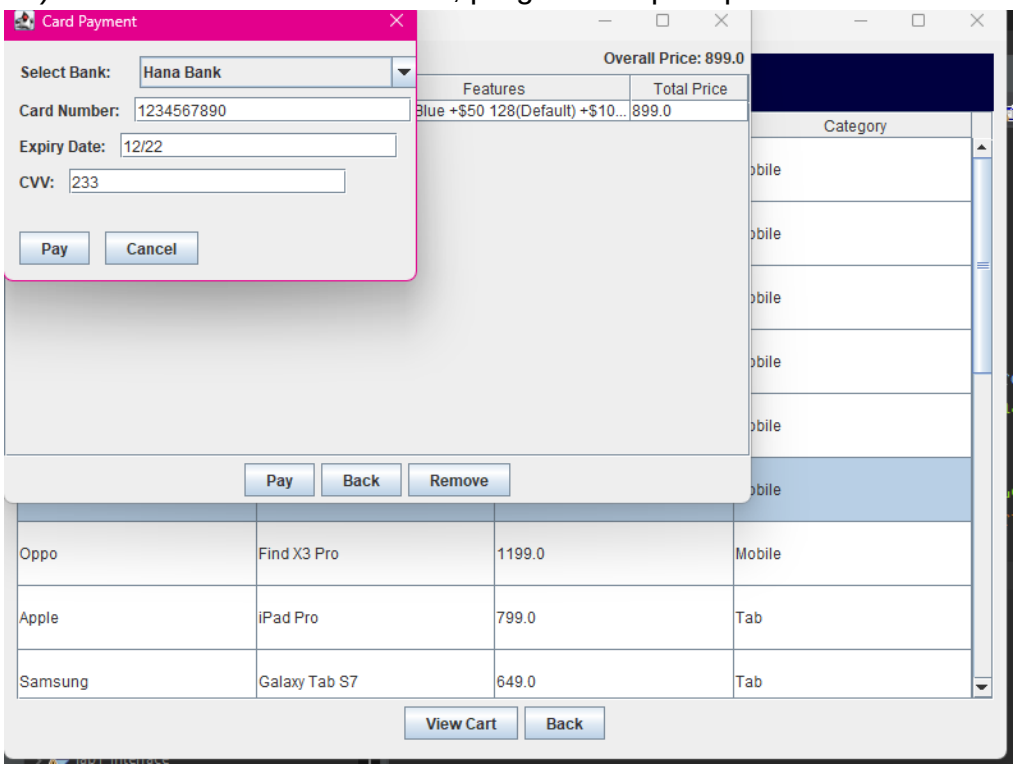
11) To remove product, click on a product and press the remove button. Cart list is then updated, overall price is updated too



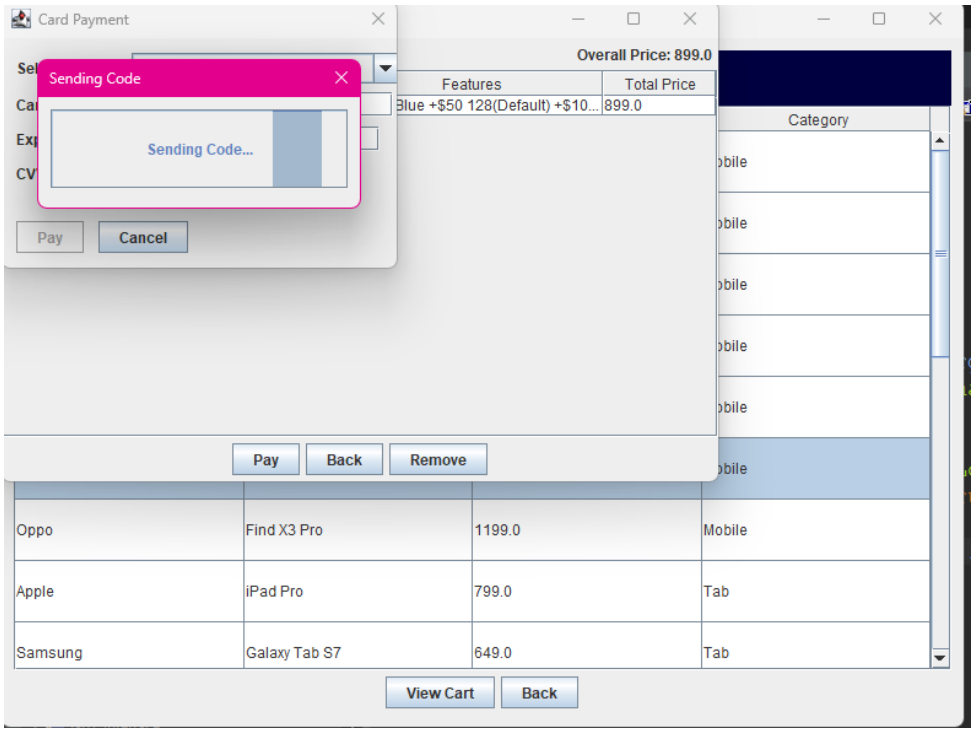
12) When you press 'Pay' button, program will ask for shipping details



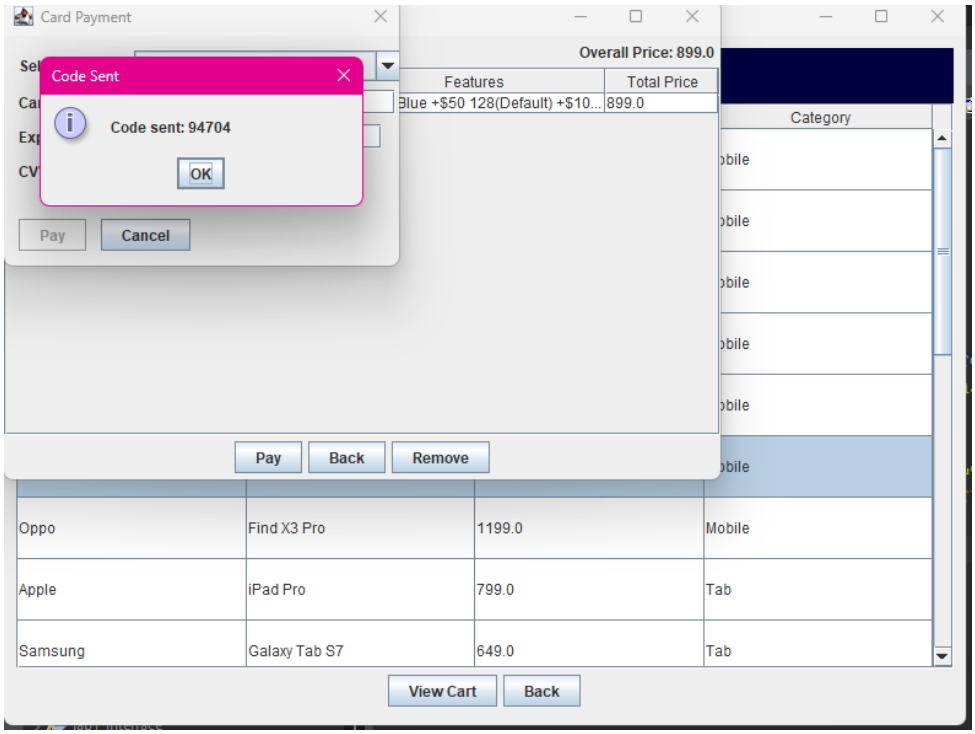
13)After 'Enter' button is clicked, program will prompt for card's information



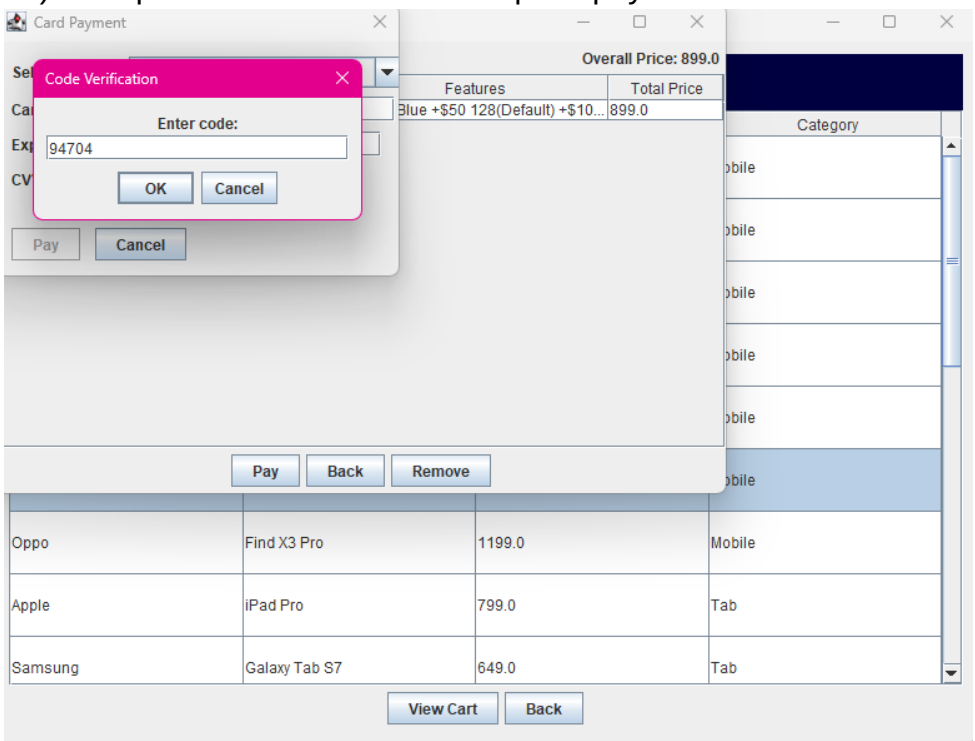
14) When user clicked 'Pay' button, program will show this progress bar indicating a code is being sent



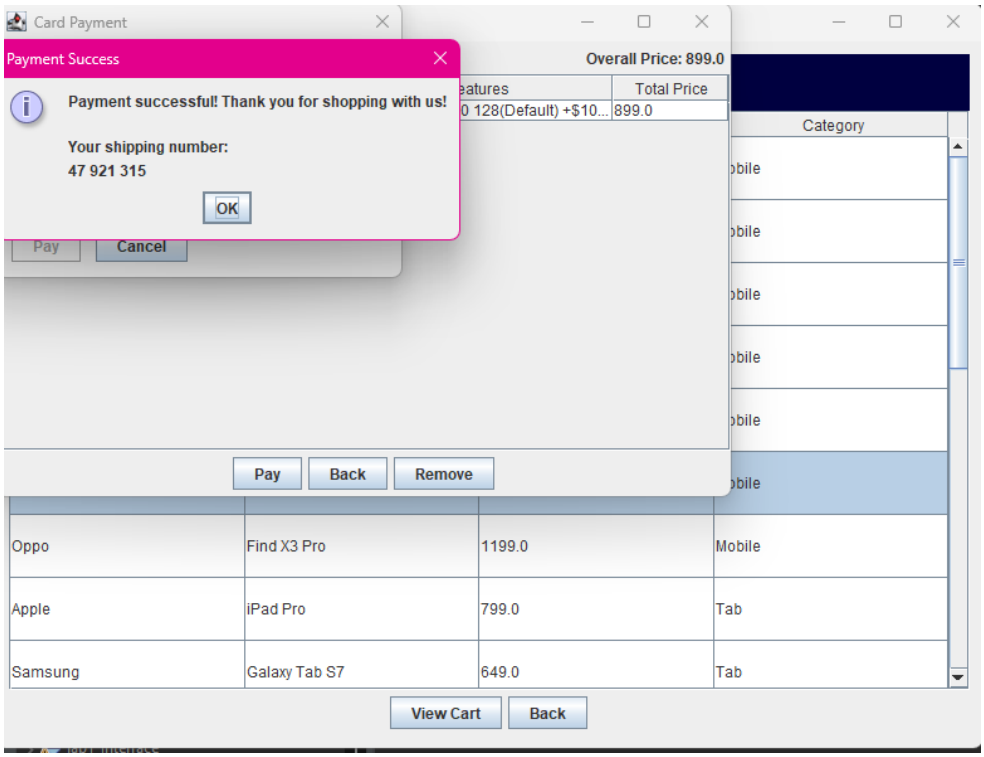
15) After 3 seconds, the confirmation is sent to complete payment. The code is randomly generated.



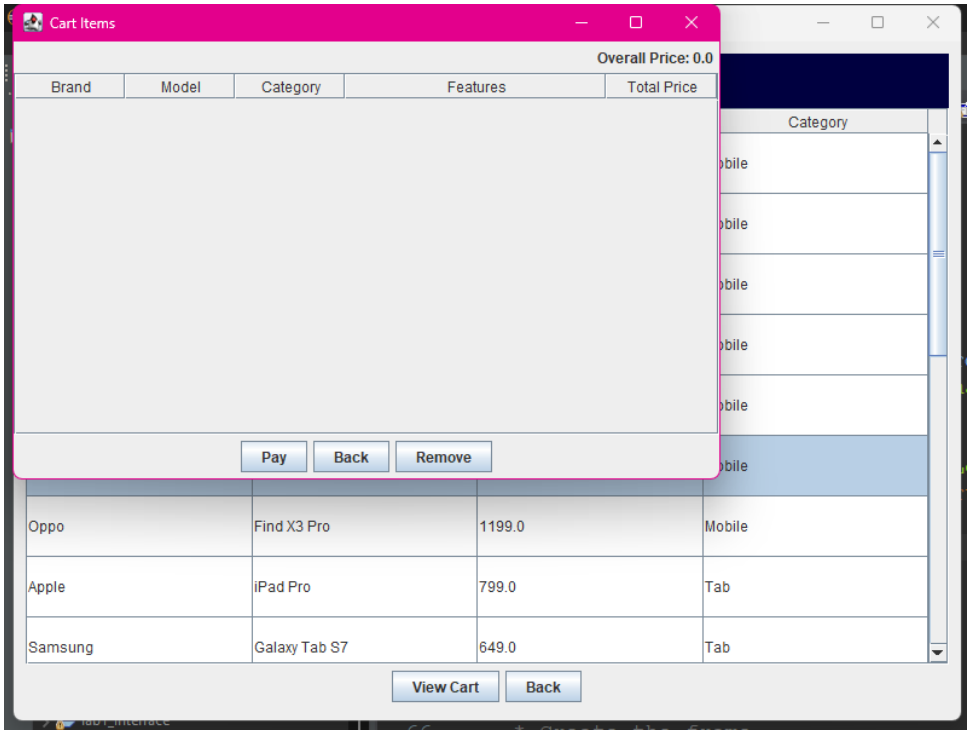
16) Prompt for the sent code to complete payment



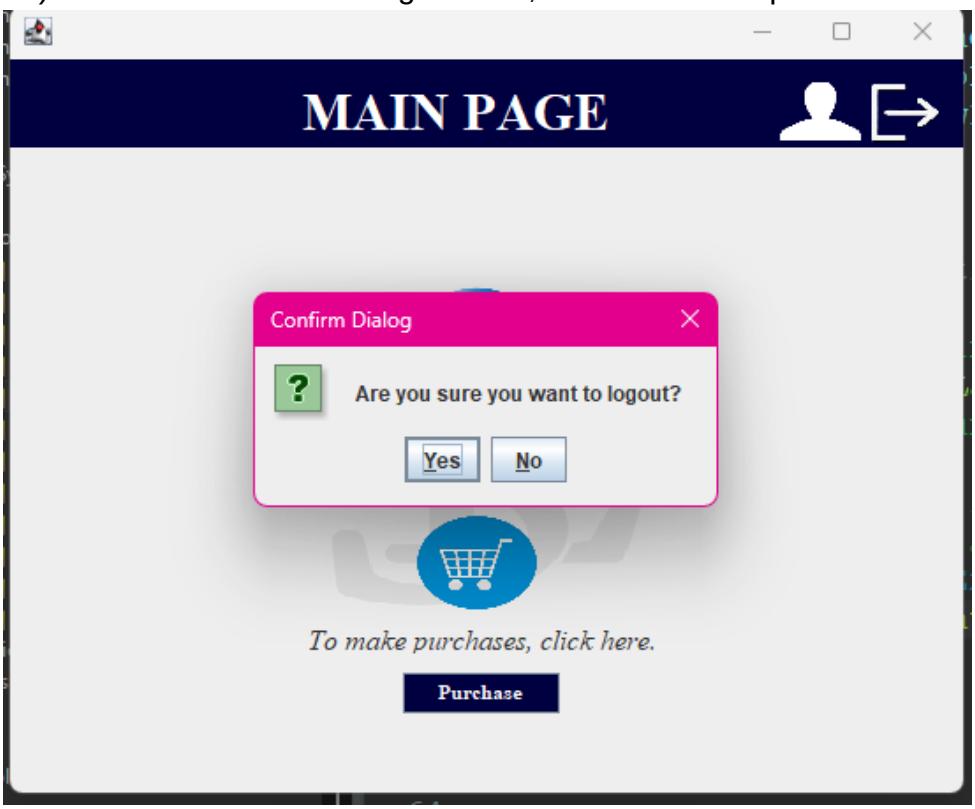
17) When payment is successful, it shows the randomly generated shipping numbers for the user



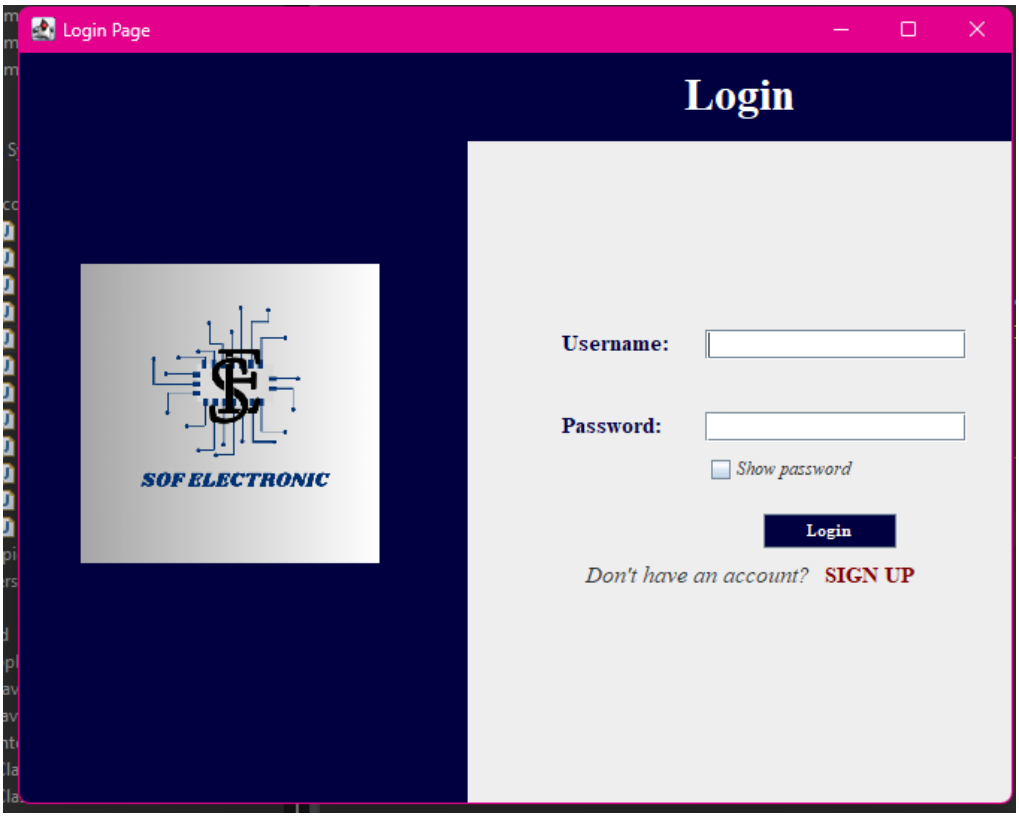
18) After payment is made, cart is emptied.



19) When user clicked on logout icon, it wil show this question



20) After logged out, user is redirected back to 'Login' page



4. Explain the code of the main functionalities

-Most of the code are using Jtable where it stored the product's information.
-One of the main functionalities would be the 'CartFrame' where all calculations are made.
-All of user prompts used Exceptions to validate user's input

5. Explain what is included in your project and why it is used (Polymorphism, Inheritance, File I/O, etc)

Polymorphism: The Timer class is used to schedule and execute tasks at specified time intervals. It demonstrates polymorphism through method overriding. In the code, a Timer object is created, and the start method is called to start the timer. The start method is inherited from the Timer class and is overridden to define the specific behavior of the timer.

Inheritance: In the code, the Product class serves as the base class for representing various products. It defines common properties and methods for products, such as brand, model, price, photo, description, category, and methods to get and set these properties.

The ProductModel class contains methods for retrieving lists of products categorized as mobile, tab, or laptop. It creates instances of the Product class and populates them with specific values for each product.
Thread: When the timer expires, it triggers an event on its dedicated thread, which executes the actionPerformed method of the ActionListener. In this case, the actionPerformed method closes the progress dialog and displays the code dialog by invoking the generateCodeDialog method.

By using a Timer thread, the UI remains responsive during the 3-second interval, and the code dialog is displayed without blocking the UI.

File I/O:
Saving User Data -The saveUserData() method is responsible for saving the user's information to a file named "users.txt". When the user successfully registers, their name, username, and password are written to the file. The method uses FileWriter and PrintWriter classes to write the data to the file in an append mode, allowing new user data to be added without overwriting existing data.
Checking for Existing Usernames: The isUsernameExists() method checks whether a given username already exists in the "users.txt" file. It reads the file line by line using a BufferedReader and checks each line to find a match with the provided username. If a match is found, it means that the username already exists. The method reads each line from the file and splits it using the colon (":") as a delimiter. It then checks if the line has two tokens (i.e., the key-value pair format "Username:username") and compares the provided username with the value in the

line. If a match is found, the method returns true, indicating that the username already exists. If no match is found, it returns false.

You can add more sections based on your project.